Arzelà–Ascoli Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of arzelà–ascoli theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

RA-05 / Real Analysis / Report v1

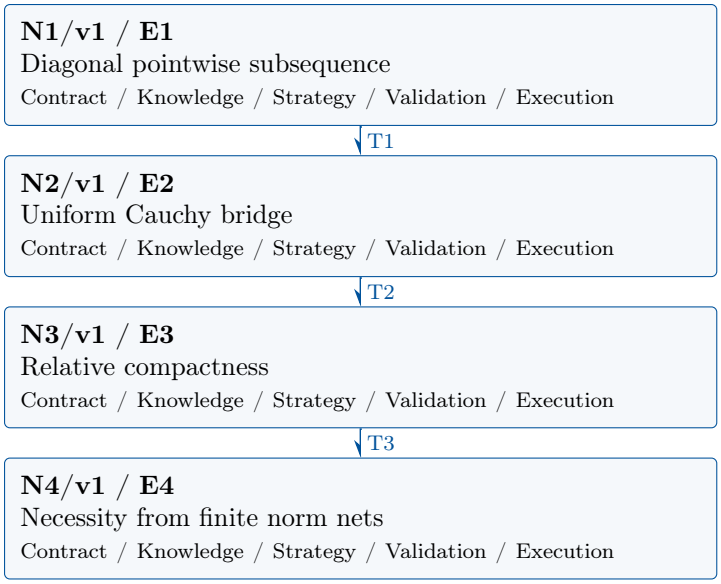
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For compact metric K, prove that $\mathcal{F} \subset C(K, \mathbb{R})$ is relatively compact in the supremum norm if and only if it is uniformly bounded and equicontinuous.*

A. Foundations and permitted imports

For compact metric K, finite ε-nets exist. Bolzano–Weierstrass applies to bounded real sequences. $C(K)$ with supremum norm is complete; a uniform limit of continuous functions is continuous. In metric spaces compactness equals sequential compactness, and complete totally bounded spaces are compact. These elementary metric lemmas are permitted imports.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A subsequence converges at all points of a countable dense set.

Receiving activity. N2: Uniform Cauchy bridge

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Diagonal convergence; uniform equicontinuity; prior exports E1/v1

T2: N2/E2/v1 to N3. Every sequence in F has a uniformly convergent subsequence in $C(K)$.

Receiving activity. N3: Relative compactness

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Subsequence property; prior exports E2/v1

T3: N3/E3/v1 to N4. Uniform boundedness and equicontinuity imply relative compactness.

Receiving activity. N4: Necessity from finite norm nets

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Compact closure(F); prior export E3/v1 for the final equivalence synthesis

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Diagonal pointwise subsequence

Object and role. Exploit a countable dense set supplied by finite nets.

Inputs. Uniform boundedness and equicontinuity of F; foundational imports stated in §1

C — Contract

Own E1/v1: A subsequence converges at all points of a countable dense set.

K — Knowledge

Retain Uniform boundedness and equicontinuity of F; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Exploit a countable dense set supplied by finite nets.

V — Validation

A dense set alone does not imply uniform convergence.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Assume K nonempty; if it is empty the function space has one element.

N1.2 Unite finite1/j-nets to obtain a countable dense $set\{z_1, z_2, \ldots\}$.

N1.3 Given any sequence in F, repeatedly extract subsequences converging at z_j using boundedness and Bolzano–Weierstrass.

N1.4 The diagonal subsequence g_j converges at every fixed z_i.

Export E1/v1. A subsequence converges at all points of a countable dense set.

Acceptance gate. A dense set alone does not imply uniform convergence. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: A dense set alone does not imply uniform convergence. Recheck dependent nodes N2, N3, N4.

B. N2/v1: Uniform Cauchy bridge

Object and role. Control the entire space by finitely many nearby sampled points.

Inputs. Diagonal convergence; uniform equicontinuity; prior exports E1/v1

C — Contract

Own E2/v1: Every sequence in F has a uniformly convergent subsequence in $C(K)$.

K — Knowledge

Retain Diagonal convergence; uniform equicontinuity; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Control the entire space by finitely many nearby sampled points.

V — Validation

The δ is common to the family, not chosen separately for each function.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Given $\varepsilon > 0$ choose δ so $d(x, y) < \delta$ implies $|h(x) - h(y)| < \varepsilon/3$ for every $h \in F$.

N2.2 Choose a finite δ-net whose centers lie in the dense set.

N2.3 For large j,k all finitely many center differences $|g_j(z_i) - g_k(z_i)|$ are below $\varepsilon/3$.

N2.4 For arbitrary x choose a center within δ; the triangle inequality gives $|g_j(x) - g_k(x)| < \varepsilon$.

N2.5 Completeness of $C(K)$ supplies a uniform continuous limit.

N2.6 Every sequence in F has such a subsequence.

Export E2/v1. Every sequence in F has a uniformly convergent subsequence in $C(K)$.

Acceptance gate. The δ is common to the family, not chosen separately for each function. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: The δ is common to the family, not chosen separately for each function. Recheck dependent nodes N3, N4.

C. N3/v1: Relative compactness

Object and role. Extend the property to the closure.

Inputs. Subsequence property; prior exports E2/v1

Contract

Own E3/v1: Uniform boundedness and equicontinuity imply relative compactness.

Knowledge

Retain Subsequence property; prior exports E2/v1; keep their assumptions and versions attached.

Strategy

Extend the property to the closure.

Validation

A limit need not belong to F itself, only its closure.

Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 For a sequence h_j in the closure of F choose $g_j \in F$ with $\|h_j - g_j\|\infty < 1/j$.

N3.2 A uniformly convergent subsequence of g_j forces the corresponding h_j to converge to the same limit, which lies in the closed set closure(F).

N3.3 Its closure is therefore sequentially compact and hence compact by the metric lemma.

N3.4 This proves sufficiency.

Export E3/v1. Uniform boundedness and equicontinuity imply relative compactness.

Acceptance gate. A limit need not belong to F itself, only its closure. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: A limit need not belong to F itself, only its closure. Recheck dependent nodes N4.

D. N4/v1: Necessity from finite norm nets

Object and role. Turn compactness into uniform bounds and a shared continuity modulus.

Inputs. Compact closure(F); prior export E3/v1 for the final equivalence synthesis

C — Contract

Own E4/v1: Relative compactness is equivalent to both stated family conditions.

K — Knowledge

Retain Compact closure(F); prior export E3/v1 for the final equivalence synthesis; keep their assumptions and versions attached.

S — Strategy

Turn compactness into uniform bounds and a shared continuity modulus.

V — Validation

Pointwise boundedness is not the hypothesis used in the diagonal step unless compact equicontinuity is separately used to upgrade it.

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Cover closure(F) by finitely many norm balls of radius 1 centered at h_i.

N4.2 Then $\|f\|\infty \leq 1 + \max_i \|h_i\|\infty$.

N4.3 For $\varepsilon > 0$ choose a finite norm net of radius $\varepsilon/3$ with continuous centers h_i.

N4.4 Each center is uniformly continuous on K; choose one δ valid for all finitely many.

N4.5 For $d(x, y) < \delta$ and f within $\varepsilon/3$ of h_i, $|f(x) - f(y)| \leq 2\|f - h_i\|\infty + |h_i(x) - h_i(y)| < \varepsilon$.

N4.6 Thus F is equicontinuous; both directions close.

Export E4/v1. Relative compactness is equivalent to both stated family conditions.

Acceptance gate. Pointwise boundedness is not the hypothesis used in the diagonal step unless compact equicontinuity is separately used to upgrade it. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Pointwise boundedness is not the hypothesis used in the diagonal step unless compact equicontinuity is separately used to upgrade it. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Diagonal pointwise subsequence

1. Assume K nonempty; if it is empty the function space has one element.
2. Unite finite1/j-nets to obtain a countable dense $set\{z_1, z_2, \ldots\}$.
3. Given any sequence in F, repeatedly extract subsequences converging at z_j using boundedness and Bolzano–Weierstrass.
4. The diagonal subsequence g_j converges at every fixed z_i.

B. §4.N2 — Uniform Cauchy bridge

1. Given $\varepsilon > 0$ choose δ so $d(x, y) < \delta$ implies $|h(x) - h(y)| < \varepsilon/3$ for every $h \in F$.
2. Choose a finite δ-net whose centers lie in the dense set.
3. For large j,k all finitely many center differences $|g_j(z_i) - g_k(z_i)|$ are below $\varepsilon/3$.
4. For arbitrary x choose a center within δ; the triangle inequality gives $|g_j(x) - g_k(x)| < \varepsilon$.
5. Completeness of $C(K)$ supplies a uniform continuous limit.
6. Every sequence in F has such a subsequence.

C. §4.N3 — Relative compactness

1. For a sequence h_j in the closure of F choose $g_j \in F$ with $\|h_j - g_j\|\infty < 1/j$.
2. A uniformly convergent subsequence of g_j forces the corresponding h_j to converge to the same limit, which lies in the closed set closure(F).
3. Its closure is therefore sequentially compact and hence compact by the metric lemma.
4. This proves sufficiency.

D. §4.N4 — Necessity from finite norm nets

1. Cover closure(F) by finitely many norm balls of radius 1 centered at h_i.
2. Then $\|f\|\infty \leq 1 + \max_i \|h_i\|\infty$.
3. For $\varepsilon > 0$ choose a finite norm net of radius $\varepsilon/3$ with continuous centers h_i.

4. Each center is uniformly continuous on K; choose one δ valid for all finitely many.
5. For $d(x, y) < \delta$ and f within $\varepsilon/3$ of h_i, $|f(x) - f(y)| \leq 2\|f - h_i\|\infty + |h_i(x) - h_i(y)| < \varepsilon$.
6. Thus F is equicontinuous; both directions close.

□

V. WORKED ORCHESTRATION

The functions on[0,1] with $|f| \leq 1$ and Lipschitz constant ≤ 1 are equicontinuous, so every sequence has a uniform convergent subsequence. The proof takes common $\delta = \varepsilon/3$ and finitely many rational grid centers; the limit keeps the same bounds.

A. Boundary and hypothesis audit

The family $x \mapsto x^n$ on[0,1] is uniformly bounded but not equicontinuous at 1. Its only possible pointwise limit has a jump at 1, so no subsequence converges uniformly. Validation rejects treating boundedness as sufficient and routes the failure to the common-modulus node.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.

6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

The family $x \mapsto x^n$ on[0,1] is uniformly bounded but not equicontinuous at 1. Its only possible pointwise limit has a jump at 1, so no subsequence converges uniformly. Validation rejects treating boundedness as sufficient and routes the failure to the common-modulus node.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.